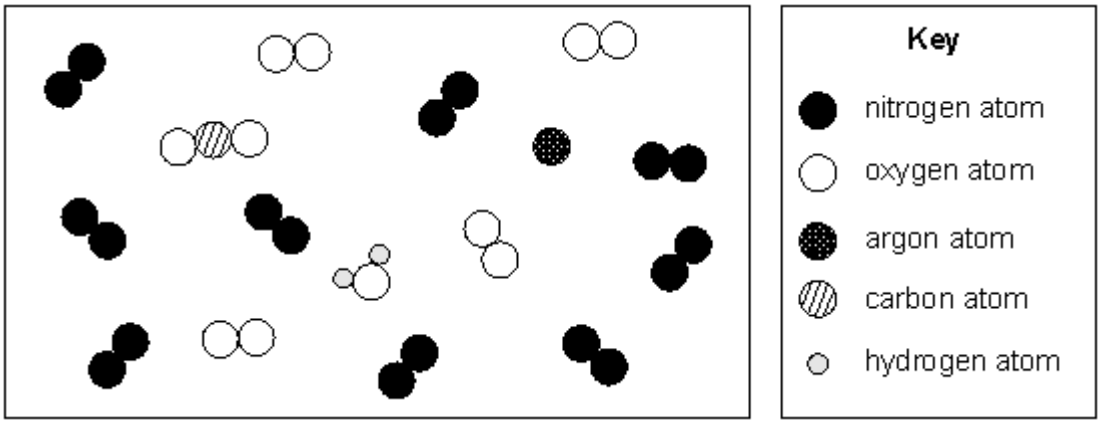
Q1.

The diagram below represents the particles found in air.



(a) Complete the following table. Use the diagram and key above to help you.

name	symbol	chemical formula
argon		Ar
nitrogen		
oxygen		O_2
		

3 marks

(b) Air is a **gas** at room temperature.
What evidence in the diagram above shows this?

.....

1 mark

(c) A sample of air in a balloon is cooled.
Complete the sentences below using words from the box.
You may use each word more than once.

increases	**decreases**	**stay the same**

When the air is cooled, the volume of the air ..... and

the mass of the air ..... .

When the air is cooled, the density of the air ..... .

1 mark

(d) In 1902, the scientist Carl von Linde cooled air to produce **liquid oxygen**.

The table below shows the melting points and boiling points of four substances that are found in air.

substance	melting point (°C)	boiling point (°C)
argon	–189	–186
oxygen	–218	–183
nitrogen	–210	–196
water	0	100

Before Linde, scientists tried to produce **liquid air** by cooling it to –190°C.
Give a reason why liquid air was not produced.

.....

.....

1 mark
maximum 6 marks

Q1.

(a) • N_2

*do **not** accept 'N2' **or** '2N' **or** 'N + N' **or** 'N^2'*

1 (L6)

• 

*do **not** accept shaded circles for O_2*
ignore size of circles

1 (L6)

• water (vapour) **or** steam H_2O

accept 'hydrogen oxide'
accept 'OH_2'
***both** answers are required for the mark*

1 (L7)

(b) • particles are separate **or** not joined **or** spread out

accept 'molecules' for 'particles'
'atoms' or 'elements' are insufficient in place of particles
'they are separate' is insufficient

1 (L7)

(c) • decreases
stays the same
increases

answers must be in the correct order
***all three** answers are required for the mark*

1 (L7)

(d) any **one** from

• water **or** argon would be solid **or** frozen

• nitrogen would be a gas

• ice would form

'air is a mixture of different gases' is insufficient

1 (L7)

[6]